Measuring Changes in Pol II Speed & Splicing Time from Total RNA-seq

Jakub Koubele

Progress Report

9th September, 2025

Introduction

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THE PREPRINT SERVER FOR BIOLOGY

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- Presentation outline:
  - Model derivation
  - Fitting and statistical inference
  - Implementation details

Model overview

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  - Several intercept terms are also included.

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- **Parameters** are the LFC of gene expression, Pol II speed, and intron splicing time.
  - Several intercept terms are also included.
- **Outputs** are the predicted exon and intron read counts, and location of reads within introns (coverage).

Total RNA-seq illustration

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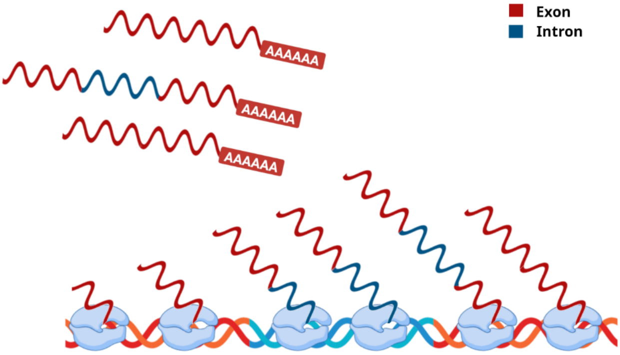

Figure 1: Intronic RNA (in blue) in total RNA-seq may come from two distinct sources: either the intron is currently being transcribed by RNA polymerase, or it was already finished, but not spliced and degraded yet. The finished introns may still be part of the nascent RNA or may come from retained introns in mature RNA.

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  - For each intron, we also record its coverage (i.e. read histogram).
- Reads not overlapping with any intron are aligned to the transcriptome, and used to construct a gene-level exon count matrix.

Modeled gene properties

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- **Expression rate** (gene-specific):

$$\eta = \eta_0 \exp\left(\mathbf{x}^\mathsf{T}\boldsymbol{\alpha}\right)$$

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- **Pol II speed** (intron-specific):

$$v = v_0 \exp\left(\mathbf{x}^{\mathsf{T}} \boldsymbol{\beta}\right)$$

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- **Splicing time** (intron-specific):

$$\tau = \tau_0 \exp\left(\mathbf{x}^\mathsf{T}\boldsymbol{\gamma}\right)$$

The unit is $[\tau] = \mathrm{s}$. The parameter $\boldsymbol{\gamma}$ is a vector with the LFC of splicing time.

Predicting exon read count

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  - Transcript abundance is also affected by the degradation rate (transcript half-life).

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- Number of reads from unspliced introns scales with the gene expression rate, and with the time needed to splice out the intron:

$$n_{\text{unspliced}} \propto \eta \tau$$

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$$\begin{aligned}\nu \propto \exp\left(\log\left(2\tau_0\eta_0\right) + \theta + \mathbf{x}^\mathsf{T}\left(\boldsymbol{\alpha} - \boldsymbol{\beta}\right)\right) \\ + \exp\left(\log\left(2\tau_0\eta_0\right) + \mathbf{x}^\mathsf{T}\left(\boldsymbol{\alpha} + \boldsymbol{\gamma}\right)\right)\end{aligned}$$

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- By introducing an intercept term, we obtain:

$$\nu = \exp\left(\text{intercept}_{\text{intron}} + \theta + \mathbf{x}^{\mathsf{T}}\left(\boldsymbol{\alpha} - \boldsymbol{\beta}\right)\right) + \exp\left(\text{intercept}_{\text{intron}} + \mathbf{x}^{\mathsf{T}}\left(\boldsymbol{\alpha} + \boldsymbol{\gamma}\right)\right)$$

Effect of Pol II speed change illustration

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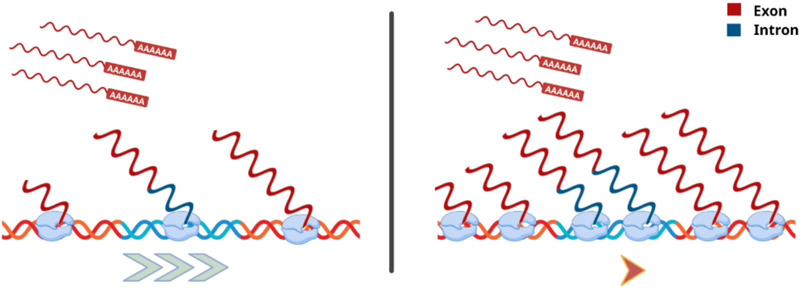

Figure 2: Effect of the change in Pol II speed. In the condition with faster polymerase (left), there is less polymerases currently transcribing a gene, compared to the condition with slower polymerase (right). Note that the amount of mature RNA is not affected by the change in elongation speed.

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- The overall distribution of intronic reads is then a mixture of the uniform and triangular components, with the density

$$p(r) = \pi p_T(r) + (1 - \pi) p_U(r) = 1 + \pi - 2\pi r, \quad r \in [0, 1]$$

Where $\pi \in [0, 1]$ is the mixture weight of the triangular distribution.

Intron coverage density illustartion

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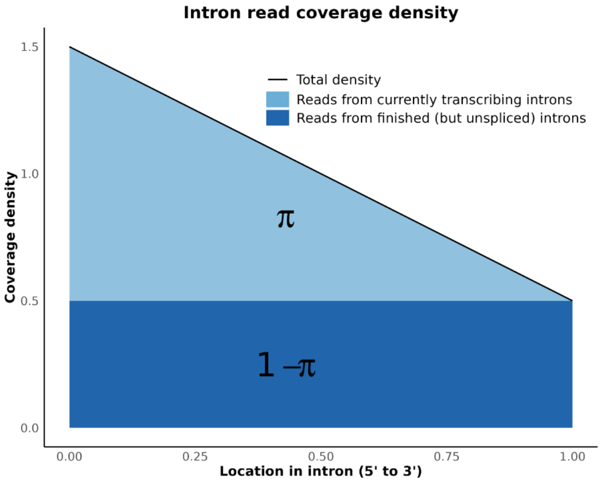

Figure 3: Density function of the intronic read location. The resulting distribution is a mixture of a decreasing triangular distribution, with a weight π, and a uniform distribution, with a weight $(1 - \pi)$.

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- The mixture parameter π depends on the number of reads from transcribing and unspliced introns:

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$$\pi = \frac{n_{\text{transcribing}}}{n_{\text{transcribing}} + 2 \cdot n_{\text{unspliced}}}$$

- After plugging in the formulas from definitions, we arrive to

$$\pi = \frac{\exp\left(\theta - \mathbf{x}^{\mathsf{T}}\left(\boldsymbol{\beta} + \boldsymbol{\gamma}\right)\right)}{1 + \exp\left(\theta - \mathbf{x}^{\mathsf{T}}\left(\boldsymbol{\beta} + \boldsymbol{\gamma}\right)\right)} = \sigma\left(\theta - \mathbf{x}^{\mathsf{T}}\left(\boldsymbol{\beta} + \boldsymbol{\gamma}\right)\right)$$

where $\sigma(.)$ denotes the logistic (sigmoid) function.

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- Optimization is performed by quasi-Newton algorithm (L-BFGS).
- Problem: the loss function is not convex (due to the reads location term).
    - However, a semi-analytical solution exists for a simple case when $\mathbf{x}$ consists of a single factor. In this case, loss minimum is unique.
    - Finding guarantees (or counter-examples) for more complex settings remains an open problem.

Hypothesis testing

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- **Likelihood-ratio test (LRT)**: Fit a restricted model (e.g., with some LFC set to 0). Difference of log-likelihood vs. unrestricted model is used in the χ^2 test.

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- **Wald test**: Assumes a Normal distribution of the parameters under H_0. The covariance matrix is estimated as the inverse of the loss Hessian.
  - Less reliable than LRT (the normality assumption is often violated), but faster to compute.

Implementation

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- Code is available on GitHub.
- Model is implemented in PyTorch.
- The whole pipeline is orchestrated by Nextflow.
- Containerization via Docker (with images hosted on Docker Hub).

The End

Thank you for your attention!

Illustrations were created using BioRender.